

"Title","Authors","Year","DOI/URL","Abstract","Source"

"How Can ChatGPT Help in Automated Building Code Compliance Checking?","Jiansong Zhang","2023","10.22260/ISARC2023/0011","One main challenge in the full automation of building code compliance checking is the extraction and transformation of building code requirements into computable representations. In this paper, the author explored the use of ChatGPT for converting building code requirements into computer code, comparing it with a semantic rule-based approach. The preliminary results indicate that while ChatGPT's conversion still has limitations compared to the rule-based approach, it shows great potential to accelerate the implementation and scale-up of automated building code compliance checking systems ¹","Proceedings of the 40th ISARC (International Symposium on Automation and Robotics in Construction) 2023"

"A GPT-based method of Automated Compliance Checking through prompt engineering","Xiaoyu Liu; Haijiang Li; Xiaofeng Zhu","2023","https://orca.cardiff.ac.uk/id/eprint/159694","This research introduces an early application of Generative Pre-trained Transformer (GPT) models for Automated Compliance Checking (ACC) in the AEC industry, requiring no additional domain knowledge or term explanation. The method involves directly inputting building design specifications and corresponding codes into the model, guided by a task-specific prompt, after which GPT generates compliance results. Initial tests on an artificially generated dataset achieved up to 91% accuracy, demonstrating the model's effectiveness. As one of the first attempts to leverage Large Language Models (LLMs) for AEC challenges, this work provides a practical workflow and dataset for future GPT-based ACC research ²","30th European Group for Intelligent Computing in Engineering (EG-ICE 2023) Conference"

"Towards Automated Regulatory Compliance Verification in Financial Auditing with Large Language Models","Armin Berger; Lars Hillebrand; David Leonhard; Tobias Deußer; Thiago B. F. de Oliveira; Tim Dilmaghani; Mohamed Khaled; Bernd Kliem; Rüdiger Loitz; Christian Bauckhage; Rafet Sifa","2023","10.1109/BigData59044.2023.10386518","Financial document auditing is traditionally labor-intensive. This paper investigates the effectiveness of Large Language Models (LLMs) in automating regulatory compliance checks in financial auditing. The study compares cutting-edge open-source LLMs (e.g., Llama-2) with proprietary models (e.g., GPT-4) using two custom datasets from PricewaterhouseCoopers (PwC) Germany. Results show that the open-source Llama-2 (70B) excels at detecting non-compliance (true negatives), outperforming proprietary counterparts, while GPT-4 performs best across a broad range of scenarios, especially in non-English contexts ³","2023 IEEE International Conference on Big Data (BigData 2023) Proceedings"

"From Automatic to Autonomous: A Large Language Model-driven Approach for Generic Building Compliance Checking","Huaquan Ying; Rafael Sacks","2024","https://easychair.org/publications/preprint/k2PJ/open","Despite decades of development in automated compliance checking (ACC) for building designs, existing systems are limited to specific domains and rule types. The diversity and complexity of design requirements demand an

intelligent, autonomous system capable of independent planning and decision-making. Inspired by the capabilities of Large Language Models (LLMs) to handle complex tasks, this study proposes an LLM-based agent for building compliance checking. The LLM agent interprets design requirements, plans the checking process, retrieves relevant BIM data, and executes checks autonomously. A proof-of-concept prototype demonstrated the potential of this LLM-driven approach to achieve more generic and scalable ACC solutions ⁴ .", "CIB W78 2024 Conference (Proc. 2024 International Conference on ICT in AEC)"

"Large Language Model-Driven Code Compliance Checking in Building Information Modeling", "Soumya Madireddy; Lu Gao; Zia Ud Din; Kinam Kim; Ahmed Senouci; Zhe Han; Yunpeng Zhang", "2025", "10.3390/electronics14112146", "This study addresses the time-consuming and error-prone nature of manual building code compliance checking by introducing an approach that integrates Large Language Models (LLMs) with BIM software (Autodesk Revit). The system uses LLMs (e.g., ChatGPT, Claude, Gemini, Llama) to interpret building codes, generate Python scripts, and perform semi-automated compliance checks within the BIM environment. Case studies on a residential house and an office building show that the LLM-driven system reduces the time and effort for compliance checks while improving accuracy. It automates identification of code violations (like non-compliant room dimensions, materials, and placements) and produces actionable reports. Compared to manual methods, the approach eliminates repetitive tasks, simplifies complex regulations, and reliably ensures adherence to standards ⁵ .", "Electronics (MDPI), Volume 14, Issue 11, 2025"

"Safeguarding Decentralized Social Media: LLM Agents for Automating Community Rule Compliance", "Lucio La Cava; Andrea Tagarelli", "2025", "10.1016/j.osnem.2025.100319", "Moderating decentralized social networks is challenging due to heterogeneous communities and vast content. This paper evaluates six AI agents built on open LLMs for automatically checking posts against community rules in decentralized social media (e.g., Mastodon). Analyzing over 50,000 posts across hundreds of servers, the LLM-based agents effectively detect non-compliant content, understand subtle language cues, and adapt to different community contexts. Most agents show high agreement with human moderators in identifying violations and justifying decisions, suggesting these LLM agents are promising tools for semi-automated, human-in-the-loop content moderation systems ⁶ .", "Online Social Networks and Media, Vol. 48 (Sept 2025)"

"From Complexity to Clarity: AI/NLP's Role in Regulatory Compliance", "Jivitesh Jain; Nivedhitha Dhanasekaran; Mona T. Diab", "2025", "https://aclanthology.org/2025.findings-acl.1366", "This survey examines the state of automated data compliance and how NLP, especially large language models, can help address its challenges. Regulatory data compliance is increasingly critical yet complex. Recent NLP advances (particularly LLMs) offer new solutions for automating compliance tasks—like analyzing privacy policies, verifying systems against regulations, and QA on legal rules. We review key challenges (adaptability, completeness, trust) and discuss how LLMs and standardized evaluation frameworks could overcome current limitations. Future research directions are proposed to improve data compliance automation, emphasizing human-AI collaboration and robust evaluation ⁷ .", "Findings of the ACL 2025 (Association for Computational Linguistics)"

"Building the Blueprint for AI-Powered Compliance Checking: Analyzing

ChatGPT-4 & Gemini by Question Category in Engineering Regulations", "Entesar Al Nama; Maqsood Mahmud", "2025", "10.1007/978-3-031-87364-5_25", "This paper explores the use of question-and-answer (Q&A) models, leveraging LLMs (ChatGPT-4 and Google Gemini), to improve access to and understanding of engineering regulations in Bahrain. The authors compare ChatGPT-4 and Gemini across five types of questions (factual, critical, hypothetical, open-ended, comparison) related to a national engineering law. Both LLMs performed well on factual queries (ChatGPT-4: ~94% correct; Gemini: ~89%) and hypothetical scenarios, with ChatGPT-4 generally achieving higher accuracy across categories (and 100% on open-ended questions). However, both models struggled with fully answering comparison questions. The findings suggest fine-tuning ChatGPT-4 could further assist engineers in navigating complex regulatory requirements ⁸ ⁹ .", "Advances in IT in Civil and Building Engineering (Proc. ICCCBE 2024), Lecture Notes in Civil Engineering vol. 629, Springer"

"A Compliance Checking Framework Based on Retrieval Augmented Generation", "Jingyun Sun; Zhongze Luo; Yang Li", "2025", "https://aclanthology.org/2025.coling-main.178", "Text-based compliance checking verifies if a company's business processes comply with relevant laws and standards using NLP. Prior approaches either use explicit logic (precise but inflexible) or semantic embeddings (flexible but losing structure). To get the best of both, this work proposes a Retrieval-Augmented Generation (RAG) framework that combines a static knowledge layer, a dynamic regulatory/business process layer, and a computational retrieval-reasoning layer. Regulatory documents are structured using an event-centric graph (focusing on actions and states rather than just entities). Experiments on Chinese and English compliance datasets show that the RAG-based framework, which incorporates structured knowledge and LLMs, achieves state-of-the-art performance, outperforming baseline methods ¹⁰ ¹¹ .", "Proceedings of the 31st International Conference on Computational Linguistics (COLING 2024), 2025"

"AI-Driven Regulatory Compliance: Transforming Financial Oversight through Large Language Models and Automation", "Hariharan Pappil Kothandapani", "2025", "https://emergingpub.com/index.php/sr/article/view/48", "This article discusses how AI, especially LLMs, is reshaping financial regulatory compliance. It highlights key AI-driven innovations such as real-time transaction monitoring (for fraud detection), automated regulatory updates via NLP, and text-to-code translation of regulations into executable rules. By leveraging these, financial institutions can improve accuracy, reduce manual effort, and stay agile with evolving rules. The paper also addresses challenges like explainability, algorithmic bias, and integration issues with legacy systems. It concludes that a balanced approach—combining advanced AI with proper oversight—is needed to build robust, transparent, and effective compliance systems in finance ¹² ¹³ .", "Emerging Science Research, Vol. 3, January 2025, pp.12-24"

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